

Retail Sales Capstone – Data Analyst Project

Objective

- Performed end-to-end Analysis of Retail Sales data to generate actionable business insights and support data-driven decision-making
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Tools & Technologies

- Excel
 - Python (Pandas, Jupyter Notebook)
 - SQL Server
 - Power BI (DAX, Power Query)
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Key Actions

- Cleaned and preprocessed multiple CSV datasets (handled nulls, duplicates, inconsistencies)
 - Designed structured data pipeline with organized project architecture
 - Loaded and analyzed data using SQL to answer key business questions
 - Built data model in Power BI with relationships and transformations
 - Developed interactive dashboards for sales, customers, products, returns, and store performance
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Key Insights

- Identified strong sales growth trends and channel dominance (in-store vs online)
- Discovered key customer segment (age 26–45) driving majority revenue
- Analyzed return patterns impacting profitability
- Highlighted store-level inefficiencies (high cost vs low profit)

Business Impact

- Enabled optimization of product strategy and store performance
- Provided actionable insights to reduce returns and improve efficiency
- Delivered dashboards for real-time monitoring and decision support

Outcome

- Built a complete **end-to-end analytics solution** from raw data to business insights
- Demonstrated strong skills in **data cleaning, modeling, analysis, and visualization**

File structure explanation

The project is organized in a structured and scalable folder hierarchy to ensure clarity and maintainability throughout the data analysis workflow. At the top level, the **Retail_Sales_Capstone** directory contains all project components. The **data** folder is divided into **raw** and **cleaned** subfolders, where raw CSV files represent the original datasets and cleaned files contain processed, analysis-ready data. The **notebooks** folder stores Jupyter Notebook files used for data cleaning and exploratory data analysis. The **sql** folder includes SQL scripts for solving business problems along with the ER diagram that defines table relationships. The **dashboard** folder contains the Power BI report file used for visualization and insights generation. The **docs** folder holds supporting documentation such as the final report and data cleaning steps. This structured approach ensures a clear separation between raw data, processed data, analysis, and reporting, making the project easy to navigate, reproduce, and extend.